ZCAS

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CAS5831 – Data Management and Processing

FINAL EXAMINATION

Date: 3rd JUNE 2026

Instructions: Answer **Question 1** and **ANY TWO Questions** from Section B

SECTION A – COMPULSORY

QUESTION ONE

The President of a book wholesaler has told you that she wants information about publishers, authors, and books. Part of the data model to help capture this information is shown in Figure 1.

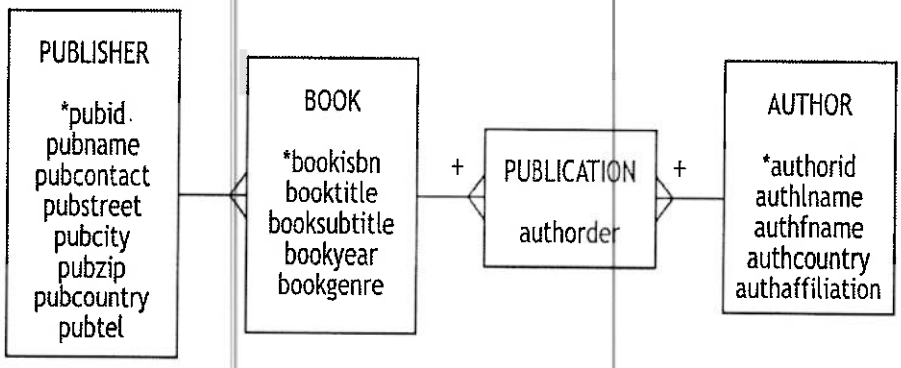


Figure 1: Part of the ER Diagram of the Book Wholesaler

(a) By carefully analyzing Figure 1, state some of the business rules that are captured in the diagram.

[10 marks]

(b) The ERD shown in figure 1 is earmarked for implementation in a Relational Database Management System such as Oracle, MySQL, etc.

Required:

Discuss the concept of a Data Model and explain why DBMSs such as Oracle are referred to as Relational DBMSs.

[10 marks]

(c) Using Relational Data Model constructs, list the set of metadata that you would use as a basis for creating the database structures of the book wholesaler database.

[15 marks]

(d) Using the Standard Structured Query Language (SQL), write SQL statements that can be used to implement the book wholesaler database model shown in Figure 1. Your statements should show all constraints as table-level constraints for ease of data management.

[15 marks]
(Total 50 marks)

SECTION B FOLLOWS ON THE NEXT PAGE

SECTION B: Attempt TWO questions from this section

Question TWO

(a) What defines ***'big data,'*** and what are its main characteristics?

[10 marks]

(b) How does the evolution of data technology influence future data management strategies?

[15 marks]

(Total 25 marks)

Question THREE

(a) By clearly explaining their role in Data Management, discuss the concepts of Data Quality, Data Governance and Data Stewardship.

[15 marks]

(b) By looking at Data Management in large organizations, such as Zambia Revenue Authority of NAPSA, outline some of the notable examples of positive data-driven decision-making?

[10 marks]

(Total 25 marks)

Question FOUR

(a) Data sharing plays a crucial role in Research and Innovation. Discuss the important role that data sharing plays in Research and Innovation.

[10 marks]

(b) What are the legal and ethical considerations in data sharing?

[15 marks]

(Total 50 marks)

END OF CAS5831 FINAL EXAM JUNE 2026